Tree Plus One-Loop Equations for the Quartic Oscillator

1 Euclidean one-loop action

Consider the Euclidean quartic oscillator

$$S_E[q] = \int \mathrm{d}\tau \left[\frac{m}{2}\dot{q}^2 + \frac{1}{2}m\omega^2 q^2 + \frac{\lambda}{24}q^4\right]. \tag{1}$$

Split

$$q(\tau) = Q(\tau) + \eta(\tau). \tag{2}$$

Expanding around an arbitrary background, and assuming boundary terms vanish, gives

$$S_E[Q+\eta] = S_E[Q] + \int \mathrm{d}\tau\, \eta(\tau) E_Q(\tau) + \frac{1}{2}\int \mathrm{d}\tau\, \eta\, \mathcal{O}_Q\, \eta + \cdots, \tag{3}$$

where

$$E_Q = -m\ddot{Q} + m\omega^2 Q + \frac{\lambda}{6}Q^3, \tag{4}$$

and the quadratic fluctuation operator is

$$\mathcal{O}_Q = -m\frac{\mathrm{d}^2}{\mathrm{d}\tau^2} + m\omega^2 + \frac{\lambda}{2}Q^2(\tau) = m\left[-\frac{\mathrm{d}^2}{\mathrm{d}\tau^2} + \Omega^2(\tau)\right], \tag{5}$$

with

$$\Omega^2(\tau) = \omega^2 + \frac{\lambda}{2m}Q^2(\tau). \tag{6}$$

In the effective-action construction the linear term is cancelled by the source. It also vanishes if the background is chosen to obey the classical equation. The one-loop effective action is

$$\Gamma_E[Q] = S_E[Q] + \frac{\hbar}{2}\operatorname{Tr}\ln \mathcal{O}_Q + O(\hbar^2). \tag{7}$$

The factor of m in $\mathcal{O}_Q$ gives only a field-independent determinant and will be dropped below.

After subtracting the field-independent $Q = 0$ determinant, the derivative expansion of the determinant is

$$\Delta\Gamma_E^{1-\mathrm{loop}}[Q] = \hbar \int \mathrm{d}\tau \left[\frac{1}{2}(\Omega - \omega) + \frac{\dot{\Omega}^2}{16\Omega^3} + O(\partial_\tau^4)\right]. \tag{8}$$

Since

$$\dot{\Omega} = \frac{\lambda Q\dot{Q}}{2m\Omega}, \tag{9}$$

we obtain the local slow-background action

$$\Gamma_E[Q] \simeq \int \mathrm{d}\tau \left[\frac{m}{2}\dot{Q}^2 + V_{\mathrm{eff}}(Q) + \hbar\,\frac{\lambda^2 Q^2 \dot{Q}^2}{64 m^2 \Omega^5}\right], \tag{10}$$

where

$$V_{\mathrm{eff}}(Q) = \frac{1}{2}m\omega^2 Q^2 + \frac{\lambda}{24}Q^4 + \frac{\hbar}{2}[\Omega(Q) - \omega], \qquad \Omega(Q) = \sqrt{\omega^2 + \frac{\lambda}{2m}Q^2}. \tag{11}$$

This local action is valid in the adiabatic regime

$$\frac{|\dot{\Omega}|}{\Omega^2} \ll 1, \qquad \frac{|\ddot{\Omega}|}{\Omega^3} \ll 1, \qquad \ldots \tag{12}$$

2 Euclidean equation of motion

Write the local action as

$$\Gamma_E[Q] \simeq \int \mathrm{d}\tau \left[\frac{1}{2}Z(Q)\dot{Q}^2 + V_{\text{eff}}(Q)\right], \tag{13}$$

with

$$Z(Q) = m + \hbar\frac{\lambda^2 Q^2}{32m^2\Omega^5}. \tag{14}$$

Varying gives

$$\frac{\partial L_E}{\partial Q} = \frac{1}{2}Z'(Q)\dot{Q}^2 + V'_{\text{eff}}(Q), \tag{15}$$

$$\frac{\partial L_E}{\partial \dot{Q}} = Z(Q)\dot{Q}, \tag{16}$$

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\frac{\partial L_E}{\partial \dot{Q}} = Z(Q)\ddot{Q} + Z'(Q)\dot{Q}^2. \tag{17}$$

Therefore the Euclidean equation of motion is

$$\boxed{-Z(Q)\ddot{Q} - \frac{1}{2}Z'(Q)\dot{Q}^2 + V'_{\text{eff}}(Q) = 0} \tag{18}$$

or equivalently

$$\boxed{Z(Q)\ddot{Q} + \frac{1}{2}Z'(Q)\dot{Q}^2 = V'_{\text{eff}}(Q).} \tag{19}$$

The needed derivatives are

$$V'_{\text{eff}}(Q) = m\omega^2 Q + \frac{\lambda}{6}Q^3 + \hbar\frac{\lambda Q}{4m\Omega}, \tag{20}$$

$$Z'(Q) = \hbar\frac{\lambda^2}{32m^2}\left[\frac{2Q}{\Omega^5} - \frac{5\lambda Q^3}{2m\Omega^7}\right]. \tag{21}$$

Thus the explicit slow-background tree plus one-loop Euclidean equation is

$$\boxed{-\left[m + \hbar\frac{\lambda^2 Q^2}{32m^2\Omega^5}\right]\ddot{Q} - \frac{\hbar\lambda^2}{64m^2}\left[\frac{2Q}{\Omega^5} - \frac{5\lambda Q^3}{2m\Omega^7}\right]\dot{Q}^2 + m\omega^2 Q + \frac{\lambda}{6}Q^3 + \hbar\frac{\lambda Q}{4m\Omega} = 0.} \tag{22}$$

Here all quantities are evaluated at $Q = Q(\tau)$, and a dot denotes $\mathrm{d}/\mathrm{d}\tau$. The equation is accurate up to $O(\hbar^2)$ and up to omitted higher-derivative terms, schematically $O(\partial_\tau^4)$ in the action.

Before the slow-background expansion, the exact Euclidean one-loop equation is nonlocal:

$$\boxed{-m\ddot{Q} + m\omega^2 Q + \frac{\lambda}{6}Q^3 + \frac{\hbar\lambda}{2}Q(\tau)G_E(\tau,\tau) = 0,} \tag{23}$$

where

$$\left[-m\frac{\mathrm{d}^2}{\mathrm{d}\tau^2} + m\omega^2 + \frac{\lambda}{2}Q^2(\tau)\right]G_E(\tau,\tau') = \delta(\tau-\tau'). \tag{24}$$

As a useful check, the diagonal Green function has the adiabatic expansion

$$G_E(\tau,\tau) = \frac{1}{m}\left[\frac{1}{2\Omega} - \frac{\ddot{\Omega}}{8\Omega^4} + \frac{3\dot{\Omega}^2}{16\Omega^5} + O(\partial_\tau^4)\right]. \tag{25}$$

Using $\dot{\Omega} = \lambda Q\dot{Q}/(2m\Omega)$, this gives

$$\frac{\hbar\lambda}{2}QG_E(\tau,\tau) = \hbar\frac{\lambda Q}{4m\Omega} - \hbar\frac{\lambda^2 Q^2}{32m^2\Omega^5}\ddot{Q} - \frac{\hbar\lambda^2}{64m^2}\left[\frac{2Q}{\Omega^5} - \frac{5\lambda Q^3}{2m\Omega^7}\right]\dot{Q}^2 + O(\partial_\tau^4). \tag{26}$$

Thus the local equation (22) is the derivative expansion of this nonlocal one-loop result.

3 Real-time one-loop equations

The real-time classical action is

$$S[Q] = \int \mathrm{d}t \left[\frac{m}{2}\dot{Q}^2 - \frac{1}{2}m\omega^2 Q^2 - \frac{\lambda}{24}Q^4\right], \tag{27}$$

where now a dot denotes $\mathrm{d}/\mathrm{d}t$. The quadratic fluctuation operator is

$$D_Q = -m\frac{\mathrm{d}^2}{\mathrm{d}t^2} - m\omega^2 - \frac{\lambda}{2}Q^2(t) + \mathrm{i}0. \tag{28}$$

The in-out one-loop effective action is

$$\Gamma_{\mathrm{in-out}}[Q] = S[Q] + \frac{\mathrm{i}\hbar}{2}\operatorname{Tr}\ln D_Q + O(\hbar^2). \tag{29}$$

Varying it gives

$$\boxed{m\ddot{Q} + m\omega^2 Q + \frac{\lambda}{6}Q^3 + \frac{\mathrm{i}\hbar\lambda}{2}Q(t)G_{\mathrm{F}}(t,t) = 0,} \tag{30}$$

where

$$D_Q G_{\mathrm{F}}(t,t') = \delta(t-t'). \tag{31}$$

Here G_{F} denotes the inverse of D_Q; with this convention the usual Feynman two-point function is $\mathrm{i}\hbar G_{\mathrm{F}}$. For a constant background, $G_{\mathrm{F}}(t,t) = -\mathrm{i}/(2m\Omega)$, and (30) reduces to

$$m\ddot{Q} + m\omega^2 Q + \frac{\lambda}{6}Q^3 + \hbar\frac{\lambda Q}{4m\Omega} = 0. \tag{32}$$

For real causal time evolution one should use the in-in, or Schwinger–Keldysh, effective action. At one loop its equation has the same tadpole form,

$$\boxed{m\ddot{Q} + m\omega^2 Q + \frac{\lambda}{6}Q^3 + \frac{\lambda}{2}Q(t)\left\langle \eta^2(t)\right\rangle_Q = 0,} \tag{33}$$

where $\langle \eta^2(t)\rangle_Q$ includes the factor of $\hbar$ and is evaluated in the Gaussian theory with time-dependent frequency $\Omega^2(t) = \omega^2 + \lambda Q^2(t)/(2m)$. For an adiabatic vacuum, $\langle \eta^2\rangle_Q = \hbar/(2m\Omega) + \cdots$.

In the same slow-background approximation used above, the real part of the local real-time one-loop action is obtained by Wick rotating (10):

$$\Gamma_{\mathrm{RT}}[Q] \simeq \int \mathrm{d}t \left[\frac{1}{2}Z(Q)\dot{Q}^2 - V_{\mathrm{eff}}(Q)\right], \tag{34}$$

with the same functions $Z(Q)$, $V_{\mathrm{eff}}(Q)$ and $\Omega(Q)$ as above. Therefore the local real-time equation of motion is

$$\boxed{Z(Q)\ddot{Q} + \frac{1}{2}Z'(Q)\dot{Q}^2 + V'_{\mathrm{eff}}(Q) = 0.} \tag{35}$$

Explicitly,

$$\boxed{\left[m + \hbar\frac{\lambda^2 Q^2}{32m^2\Omega^5}\right]\ddot{Q} + \frac{\hbar\lambda^2}{64m^2}\left[\frac{2Q}{\Omega^5} - \frac{5\lambda Q^3}{2m\Omega^7}\right]\dot{Q}^2 + m\omega^2 Q + \frac{\lambda}{6}Q^3 + \hbar\frac{\lambda Q}{4m\Omega} = 0.} \tag{36}$$

All quantities in (36) are evaluated at $Q = Q(t)$. This equation is the real-time adiabatic, local form of the one-loop corrected equation. For strongly time-dependent backgrounds, the full real-time equation is nonlocal and, in the in-out formalism, can acquire an imaginary part describing nonadiabatic excitation of fluctuations.